REFINING A PH COVERING LOCALLY ON THE TARGET

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file topologies.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): the official pool candidate recorded statement [2663, 2673] and proof(s) [[2675, 2703]]; this spec covers the complete environment instead (statement 2663-2673, proof span(s) [(2675, 2703)]), since in this chapter the author's proof environment(s) follow the statement environment and some pool entries under-counted them.

Editorial note. One 'small detail omitted' concerns closure of the refinement class under the open-closed affine refinement bookkeeping; the construction and its inputs are written.

Source reference. Original definition 0DBG: sources/stacks-chow/tags.txt, source lines 16696–16696. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let T be a scheme. Let $\{f_i : T_i \to T\}_{i\in I}$ be a family of morphisms such that f_i is locally of finite type for all i. The following are equivalent*

(1) *$\{T_i \to T\}_{i\in I}$ is a ph covering,*
(2) *there is a ph covering which refines $\{T_i \to T\}_{i\in I}$, and*
(3) *$\{\coprod_{i\in I} T_i \to T\}$ is a ph covering.*

Proof. The equivalence of (1) and (2) follows immediately from Definition 0DBG and the fact that a refinement of a refinement is a refinement. Because of the equivalence of (1) and (2) and since $\{T_i \to T\}_{i\in I}$ refines $\{\coprod_{i\in I} T_i \to T\}$ we see that (1) implies (3). Finally, assume (3) holds. Let $U \subset T$ be an affine open and let $\{U_j \to U\}_{j=1,\ldots,m}$ be a standard ph covering which refines $\{U \times_T \coprod_{i\in I} T_i \to U\}$. This means that for each j we have a morphism

$$h_j : U_j \longrightarrow U \times_T \coprod\nolimits_{i\in I} T_i = \coprod\nolimits_{i\in I} U \times_T T_i$$

over U. Since U_j is quasi-compact, we get disjoint union decompositions $U_j = \coprod_{i\in I} U_{j,i}$ by open and closed subschemes almost all of which are empty such that $h_j|_{U_{j,i}}$ maps $U_{j,i}$ into $U \times_T T_i$. It follows that

$$\{U_{j,i} \to U\}_{j=1,\ldots,m,\ i\in I,\ U_{j,i}\neq\emptyset}$$

is a standard ph covering (small detail omitted) refining $\{U \times_T T_i \to U\}_{i\in I}$. Thus (1) holds. □

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